Attention-Aware Dynamic Self-Aggregation Network for Satellite Image Time Series Classification

Wei Zhang, Peijun Du, *Senior Member, IEEE*, Pingjie Fu, Peng Zhang, Pengfei Tang, Hongrui Zheng, Yaping Meng, and Erzhu Li

***Abstract*— An effective network structure is essential for the classification of satellite image time series (SITS). Deep learning models have been widely used for SITS classification and achieved impressive performance, especially the architectures based on self-attention. However, the lack of efficient and comprehensive attention to valuable bands and time series structure hinders the performance to some extent. To address this problem, an end-to-end attention-aware dynamic self-aggregation network (ADSN) is proposed for SITS classification in this work, which combines two main parts: spectral focusing and spectral–temporal feature learning. The core components of ADSN are the channel attention module and dynamic self-aggregation block. Specifically, informative bands in the SITS flowing through the channel attention module can adaptively get a high weight to increase their contributions, while the attentions of some low-efficiency bands are weakened. Besides, the dynamic self-aggregation block, which integrates multiscale dynamic convolution and improved multihead attention in parallel, can simultaneously capture long- and short-distance sequence structures and position relationships to better represent temporal information. Compared with random forest (RF) and seven deep learning algorithms, the proposed model effectively learns spectral and temporal features, and the experimental results confirm that ADSN has achieved superior classification accuracy and generalization ability on two SITS datasets with extremely unbalanced samples.**

***Index Terms*— Channel attention module, multihead attention mechanism, multiscale dynamic convolution, satellite image time series (SITS) classification.**

I. Introduction

FOLLOWING the launch of a series of satellites, remote sensing images provide rich data sources for Earth observation. The high-frequency, large-scale, and publicly available characteristics of satellite data provide a great convenience for the application of satellite image time series (SITS) [1]. More specifically, the Landsat program, jointly operated through NASA and USGS, has adopted an open data policy since 2008, making archived long-term and large-area observation data available for free [2]. Recently, a series of Sentinel satellites, promoted by the European Union Copernicus program, provides multitemporal (total revisit period is about two to five days), higher resolution (Sentinel-2 multispectral imagery with up to 10-m visible/near-infrared bands), and big volume (Sentinel-2 images of 7.76 TiB can be released on a daily average) land images [3], [4]. The moderate resolution SITS, such as Sentinel-2 and Landsat TM/ETM+/OLI, play an important role in crop mapping [5], land management [6], forest monitoring [7], [8], offshore wind farms recognition [9], and so on. These successful cases highlight the importance of effectively exploiting abundant data and mining useful knowledge in time series images.

In practical applications, apart from the emphasis on spatial and spectral features [10], [11], the exploration of temporal information of satellite images has also received continuous attention. The SITS classification is one of the main tasks of dynamic monitoring, and its goal is to assign class labels to pixels or parcels based on the temporal features of spectral trajectory [12]. Various methods have been developed to promote the SITS classification. One is to fully explore the breakpoints of time series and the changing trend of time segments to serve the classification task, which depends on raw reflectance, vegetation indices, phenological metrics, and their fitting and decomposition components [13], [14]. The famous algorithms involve continuous change detection and classification (CCDC) [15], LandTrendr [16], BFAST [17], and so on. However, the dependence on specific types of data prevents the generalization of such approaches to some extent. The other is traditional machine learning algorithms combined with handcrafted features, such as random forest (RF) and support vector machine [18], [19]. Although these means are interpretable, they tend to ignore the sequential structure and positional relationship in the temporal dimension. In addition, these handcrafted features are, sometimes, too undercharacterized to cope with complex situations caused by disturbances and shifts in land cover.

Recent advances in deep learning have shown that end-to-end neural networks can efficiently mine learn-

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able features of complicated data [20]–[24], which outperforms traditional algorithms. There are three main deep learning architectures for SITS classification: recurrent neural network (RNN) [25] integrating image series information by updating the hidden state at each time step, convolutional neural network (CNN) [26] aggregating fixed size sequences through multiple convolutional layers, and the self-attention network [27] directly summarizing all image sequences via parallel multihead attention mechanism.

RNN, including long short-term memory (LSTM) [5] and STAR RNN (StarRNN) [28], has been widely used in the acquisition of temporal characteristics and land cover mapping [29]. Yet, since long-term dependence requires deeper structures, the problems of vanishing and exploding gradients are prone to occur. In addition, the inability to perform parallel operations increases the time consumption. CNN seems to be the natural choice to process spatial information. Temporal 1-D CNN (TempCNN) [30] and temporal convolutional networks (TCNs) [31], where convolutions were applied in the temporal dimension, have proven to be effective for addressing SITS classification. Also, 3-D CNN is an efficient feature extractor for the representation of spatiotemporal information in the time series classification [32]. Inevitably, the fixed structure and single scale of temporal CNN may affect the applicability of the same network for image series of different lengths. To simultaneously utilize spatial and temporal information of SITS, hybrid neural networks integrating convolutional and recurrent units have been comprehensively investigated [33]–[35]. With the emergence of self-attention as a powerful alternative to RNN in the natural language processing, Rußwurm and Körner [2] proposed to adopt Transformer based on self-attention for SITS classification. Similarly, the pixel-set encoders and temporal attention encoder (PSE-TAE) introduced master query strategy to optimize Transformer for embedding time-sequences [27], [36]. Their detailed experiments demonstrate that the self-attention network allows the model to focus on some crucial observations by considering the temporal structure of input data, which enables similar architectures to achieve better classification accuracy than convolution-recurrent networks.

The model based on self-attention has been remarkably adapted at SITS classification [37]. However, there are still some issues that have not been well solved.

1) In self-attention, the attention is only distributed on the time series structure of the image bands, and different bands are not effectively filtered. Hence, how to adaptively select valuable bands as input through data driven to improve the SITS classification accuracy of the end-to-end model?
2) The self-attention mainly relies on global information in the temporal dimension of SITS to make decisions, resulting in insufficient use of local information. Therefore, if the long- and short-range sequence structures can be simultaneously captured, it will be beneficial for better representation of time series information.

Inspired by visual attention and parallel thinking, this study proposes an end-to-end attention-aware dynamic self-aggregation network (ADSN) to address the above problems for SITS classification. The key contributions of this article are given as follows.

1) The channel attention module is designed to adaptively learn the weight of each spectral band in SITS.
2) To solve the inadequate utilization of local information, dynamic self-aggregation block is proposed by integrating constructed multiscale dynamic convolution [38] and improved multihead attention mechanism [39], which takes into account the global and local information in the temporal dimension of SITS.

The rest of this article is organized as follows. Section II details the proposed model. The datasets, experimental setup, and discussion of results are presented in Section III. Finally, Section IV is the conclusion and future research outlook.

II. Methodology

A. Overview of the Proposed Model

Fig. 1 is the schematic of the proposed model, consisting of two main parts: channel attention module and spectral–temporal feature learning. The channel attention module is engaged in learning the weights of the bands and makes the model concentrate on valuable spectral information. In the spectral–temporal feature learning module, the dynamic self-aggregation block can efficiently extract and fuse the structural features of temporal dimension for SITS classification.

B. Channel Attention Module

By assigning weights to the input sequence, the attention mechanism is used to highlight or select the useful features while suppressing irrelevant details [40]. It has become increasingly popular and performs well in various sequence tasks, such as natural language processing [41], image classification [42], [43], and speech recognition [44]. In view of the lightweight channel and spatial attention designs [45], [46], the channel attention module is proposed to emphasize the meaningful bands for better feature representation and classification of SITS.

The channel attention module is shown in Fig. 2. The SITS consists of images with T time steps; let the image of the tth time step be $\boldsymbol{X}_t = \{\boldsymbol{x}_{(t,1)}, \boldsymbol{x}_{(t,2)}, \ldots, \boldsymbol{x}_{(t,c)}, \ldots, \boldsymbol{x}_{(t,C)}\} \in \mathbf{R}^{C\times H\times W}$, where the number of image bands is C and the spatial size is $H \times W$. The channel attention map is generated by using the inter-bands relationship of the image series. Average pooling is commonly used to aggregate spatial information, which helps characterize the overall statistical distribution of the target object [45], [47]. Max pooling collects crucial clues on distinctive object characteristics to infer more effective channelwise attention [46]. Therefore, the two operations are adopted simultaneously to enhance the representation ability of the network [46]. To produce diverse feature descriptors at the tth time $\boldsymbol{F}_{\text{avg_}t} \in \mathbf{R}^C$ and $\boldsymbol{F}_{\max_t} \in \mathbf{R}^C$, the global average pooling and the global max pooling are conducted respectively, as done in [46]. $\boldsymbol{F}_{\text{avg_}t}$ and $\boldsymbol{F}_{\max_t}$ of image $\boldsymbol{X}_t$

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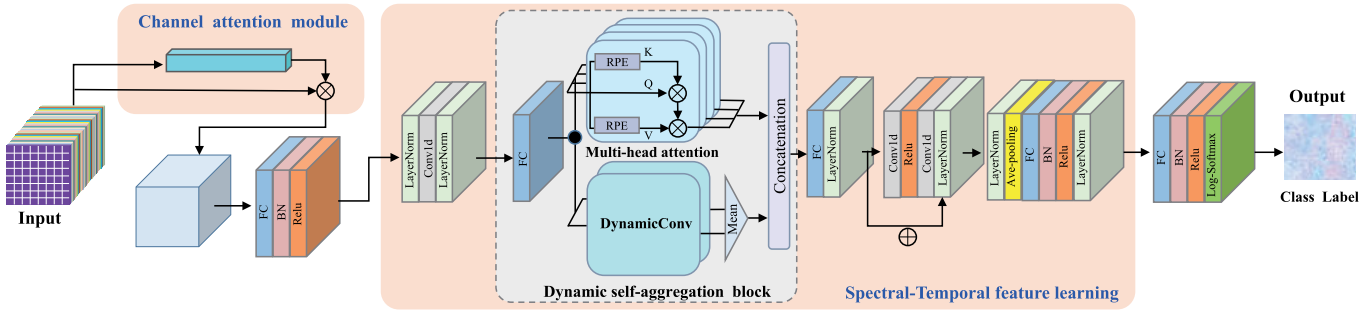


Fig. 1. Overall procedure of ADSN for SITS classification. The meanings of the abbreviations and symbols involved in the procedure are given as follows: FC: fully connected layers; BN: batchNorm1d; Conv1d: 1-D convolution; DynamicConv: dynamic convolution; RPE: relative position representation; ⊗: elementwise multiplication; ⊕: residual connection; Q: query matrix; K: key matrix; and V: value matrix.

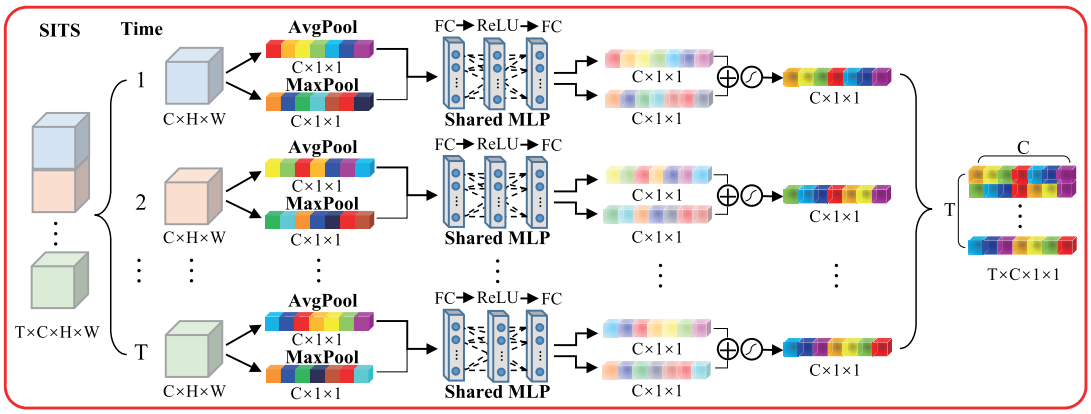


Fig. 2. Channel attention module. The model exploits both max-pooling and average-pooling outputs with a weight-shared network. Some of the abbreviations include the following: AvgPool: average pooling; MaxPool: max pooling; FC: fully connected layers; and ReLU: ReLU function [48].

can be formulated as

$$\boldsymbol{F}_{\text{avg_}t} = \left\{F_{(\text{avg_}t,1)}, F_{(\text{avg_}t,2)}, \ldots, F_{(\text{avg_}t,c)}, \ldots, F_{(\text{avg_}t,C)}\right\} \tag{1}$$

$$F_{(\text{avg_}t,c)} = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \boldsymbol{x}_{(t,c)}(i,j) \tag{2}$$

$$\boldsymbol{F}_{\text{max_}t} = \{F_{(\text{max_}t,1)}, F_{(\text{max_}t,2)}, \ldots, F_{(\text{max_}t,c)}, \ldots, F_{(\text{max_}t,C)}\} \tag{3}$$

$$F_{(\text{max_}t,c)} = \max\left(\boldsymbol{x}_{(t,c)}\right) \tag{4}$$

where t refers to the tth time step, $F_{(\text{avg_}t,c)}$ represents the result of global average pooling for the cth band, $F_{(\text{max_}t,c)}$ means the result of global max pooling for the cth band, $\boldsymbol{x}_{(t,c)}(i,j)$ denotes the variable of the cth channel $\boldsymbol{x}_{(t,c)}$ at spatial position (i,j), and max(·) refers to the maximum value of $\boldsymbol{x}_{(t,c)}$. Besides, C is the total number of bands or channels of image, and the spatial size of image is $H \times W$.

Then, the shared multilayer perceptron (MLP) is employed for information exchange, which aims to enrich the interaction between spectral bands and promote generalization. Specifically, the shared MLP is composed of fully connected (FC) layers and ReLU function [48] (see Fig. 2). It has one hidden layer, and the number of units is affected by the number of channels and reduction ratio. Generally, the simplified unit number of shared MLP can be expressed as C to C/r and then to C, where C is the number of channel and r denotes the reduction ratio. Furthermore, each feature descriptor is processed by shared MLP, and the outputs are combined to channel attention by elementwise summation. For the input at the tth time, this process can be represented by

$$\boldsymbol{S}_t = \sigma\left(\boldsymbol{W}_1\left(\text{ReLU}\left(\boldsymbol{W}\left(\boldsymbol{F}_{\text{avg_}t}\right)\right)\right) + \boldsymbol{W}_1\left(\text{ReLU}\left(\boldsymbol{W}\left(\boldsymbol{F}_{\text{max_}t}\right)\right)\right)\right) \tag{5}$$

where $\sigma(\cdot)$ is the sigmoid function, the weights are $\boldsymbol{W} \in \mathbf{R}^{C/r \times C}$ and $\boldsymbol{W}_1 \in \mathbf{R}^{C \times C/r}$, and $\boldsymbol{S}_t$ refers to channel attention, that is, $\boldsymbol{S}_t = \{\boldsymbol{s}_{(t,1)}, \boldsymbol{s}_{(t,2)}, \ldots, \boldsymbol{s}_{(t,c)}, \ldots, \boldsymbol{s}_{(t,C)}\}$. In this work, r is set 1.

After processing through the channel attention module, the result of input from the tth time step is acquired by rescaling the input features

$$\boldsymbol{H}_{(t,\text{out})} = \left\{\boldsymbol{H}_{(\text{avg_}t,1)}, \boldsymbol{H}_{(\text{avg_}t,2)}, \ldots, \boldsymbol{H}_{(\text{avg_}t,c)}, \ldots, \boldsymbol{H}_{(\text{avg_}t,C)}\right\} \tag{6}$$

$$\boldsymbol{H}_{(t,c)} = \text{F}_{\text{scale}}\left(\boldsymbol{h}_{(t,c)}, s_{(t,c)}\right) = s_{(t,c)} \cdot \boldsymbol{h}_{(t,c)} \tag{7}$$

where $\boldsymbol{H}_{(t,c)}$ is the readjustment weight sequence of different channels, $\boldsymbol{H}_{(\text{avg_}t,c)}$ denotes the cth reweighted channel, and $\text{F}_{\text{scale}}(\boldsymbol{h}_{(t,c)}, s_{(t,c)})$ means channelwise multiplication both the scaling factor $s_{(t,c)}$ and the cth channel item $\boldsymbol{h}_{(t,c)}$.

Finally, based on the operation of $\boldsymbol{H}_{(t,\text{out})}$, the reweighted SITS sequence $\boldsymbol{H}$ can be obtained, that is, $\boldsymbol{H} = \{\boldsymbol{H}_{(1,\text{out})}, \boldsymbol{H}_{(2,\text{out})}, \ldots, \boldsymbol{H}_{(t,\text{out})}, \ldots, \boldsymbol{H}_{(T,\text{out})}\}$.

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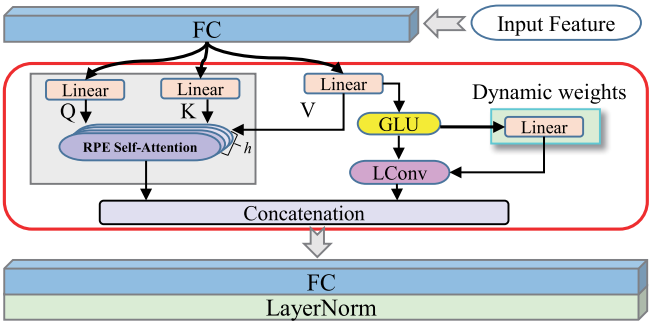


Fig. 3. Proposed dynamic self-aggregation block integrating the PRE multihead attention and dynamic convolution. Some of the abbreviations include the following: FC: fully connected layers; Q: query matrix; K: key matrix; V: value matrix; Linear: linear transformation [49]; GLU: gated linear unit [55]; and LConv: LightConv [56].

C. Dynamic Self-Aggregation Block

The proposed dynamic self-aggregation block is the core of spectral–temporal feature learning, which helps the model comprehensively find discriminative information. Based on the parallel strategy, we adopt the improved multihead attention with relative position encoding (RPE) [49] to condense the global characteristics in the temporal dimension. Dynamic convolution is introduced to supplement multihead attention by extracting local information of the time series.

More specifically, the weights of RPE multihead attention are computed by comparing the current time step to other elements in time series, which takes into account the long-range dependence of single time step element [49], [50]. Different from RPE multihead attention, dynamic convolution predicts different convolution kernels at each time step, and the kernel is only a function of the current time step without considering the entire context in time series. The weights generated by dynamic convolution help to reflect the local information and the characteristics of short-range structure on sequence [51]. Therefore, the functions of the RPE multihead attention and dynamic convolution are exactly complementary, and the hybridization of them is beneficial to capture the global and local information in the temporal dimension of SITS.

In view of the above analysis, the dynamic self-aggregation block is proposed by integrating constructed multiscale dynamic convolution and RPE multihead attention. The specific structure is shown in Fig. 3. It encodes inputs into hidden representations and then employs RPE multihead attention and multiscale dynamic convolution in parallel; results from each part are concatenated and input to the fully connected and LayerNorm layers. Dynamic convolution makes up for the lack of local information usage, while RPE multihead attention focuses on capturing long-range dependence. More importantly, this block has high expansibility, and effective transformations can be easily introduced as new parallel branches.

1) RPE Multihead Attention: Self-attention has originated from Transformer, and its main block is the multihead attention mechanism [50]. Different from CNN and RNN, Transformer usually uses absolute position encoding to perceive information on the sequential position of elements. Recently, an efficient approach incorporating RPE into self-attention has shown tremendous potential, which considers the pairwise relationship between input elements [49].

Assume that the processed SITS becomes $\boldsymbol{P} = \{\boldsymbol{\rho}_1, \boldsymbol{\rho}_2, \ldots, \boldsymbol{\rho}_i, \ldots, \boldsymbol{\rho}_T\}$ of T time steps, where $\boldsymbol{\rho}_i \in \mathrm{R}^{\mathrm{dc}}$. Relying on the input sequence, each attention head is used to construct a new sequence $\boldsymbol{Z} = \{z_1, z_2, \ldots, z_T\}$ and $z_i \in \mathrm{R}^{dz}$. Mathematically, z_i is given by linear transformation and summation

$$z_i = \sum_{j=1}^{T} \alpha_{ij}\left(\boldsymbol{\rho}_i \boldsymbol{W}^V + \boldsymbol{a}_{ij}^V\right) \tag{8}$$

$$\alpha_{ij} = \frac{\exp\left(e_{ij}\right)}{\sum_{k=1}^{T} \exp(e_{ik})} \tag{9}$$

$$e_{ij} = \frac{\boldsymbol{\rho}_i \boldsymbol{W}^Q \left(\boldsymbol{\rho}_j \boldsymbol{W}^K\right)^{\mathrm{T}} + \boldsymbol{\rho}_i \boldsymbol{W}^Q \left(\boldsymbol{a}_{ij}^K\right)^{\mathrm{T}}}{\sqrt{d_z}} \tag{10}$$

where α_{ij} is weight coefficient, $\boldsymbol{W}^Q$, $\boldsymbol{W}^K$, and $\boldsymbol{W}^V$ denote parameter matrices, and $\boldsymbol{a}_{ij}^K$ and $\boldsymbol{a}_{ij}^V$ mean edge vectors between $\boldsymbol{\rho}_i$ and $\boldsymbol{\rho}_j$. Besides, $\exp(\cdot)$ represents an exponential function based on the natural constant e, and Q, K and V indicate linear transformations for query, key, and value, respectively.

For the RPE, the edges can catch information about the relative position divergences between input elements. The upper bound of relative position is limited to the maximum absolute value m. Specifically, the focus range is $2m + 1$. Thus, edge vectors $\boldsymbol{a}_{ij}^K$ and $\boldsymbol{a}_{ij}^V$ can be formulated as

$$\boldsymbol{a}_{ij}^K = \boldsymbol{\omega}_{\mathrm{clip}(j-i,m)}^K, \quad \boldsymbol{a}_{ij}^V = \boldsymbol{\omega}_{\mathrm{clip}(j-i,m)}^V \tag{11}$$

where the clip operation is expressed as $\mathrm{clip}(x, m) = \max(-m, \min(m, x))$, $\boldsymbol{\omega}^K = \{\omega_{-m}^K, \ldots, \omega_m^K\}$, and $\boldsymbol{\omega}^V = \{\omega_{-m}^V, \ldots, \omega_m^V\}$ are relative position representations that need to be learned by the model.

2) Dynamic Convolution: Depthwise convolution (DepthConv) is suitable for learning sequence expression in hidden space [52]. Among them, lightweight convolution (LightConv) is an efficient carrier, which is depthwise separable, softmax-normalized across the temporal dimension and shares weights over the channel dimension [51], [52]. Dynamic convolution built on LightConv [51] is chosen as convolution operator in this work.

Specifically, dynamic convolution predicts different convolution kernels at each time step, which is similar to the local connection layer, and the weight is dynamically generated by the network and changes at each position [51], [53]. Each convolution submodule embodies multiple units with diverse kernel sizes [54]. They can be applied to capture features in different ranges to exert multiscale effects. After convolution units operate on the input $\boldsymbol{P}$, the output is

$$\mathrm{Conv}_k(\boldsymbol{P}) = \mathrm{Depth_conv}_k\left(\boldsymbol{P}\boldsymbol{W}^V\right)\boldsymbol{W} \tag{12}$$

where $\boldsymbol{W}^V$ is the projection transformation matrix, $\boldsymbol{W}$ refers to output parameter matrix, and $\mathrm{Depth_conv}(\cdot)$ represents depth convolution from [51]. Finally, for the input of the ith element

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in the time series, the output O is defined as

$$\boldsymbol{O}_{i,c} = \sum_{j=1}^{k}\left(\text{softmax}\left(\sum_{c=1}^{d}\boldsymbol{W}_{j,c}\boldsymbol{P}_{i,c}\right)\cdot\boldsymbol{P}_{i+j-[\frac{k+1}{2}],c}\right) \quad (13)$$

where k is the kernel size, d is the hidden size, c denotes the output channel, and $\boldsymbol{W}_{j,c}$ means the parameter matrix.

D. Auxiliary Network Layer

FC is a fully connected layer that performs a linear transformation on the input tensor. BN means applying batch normalization to the input. This operation will produce a more predictable and stable gradient, thus speeding up the training of deep neural network [57], [58]. ReLU represents the operation of rectified linear unit function [59]. Specifically, ReLU causes sparseness, reduces the interdependence between network parameters, and alleviates the occurrence of overfitting problems [48]. LayerNorm denotes the layer normalization [60], which is effective in stabilizing the hidden state dynamics in recurrent networks and can greatly reduce training time. Conv1d stands for 1-D convolution [61].

It can be seen from Fig. 1 that two combined operations are involved between the channel attention module and the dynamic self-aggregation block. The first is FC+BN+ReLU, and the second is LayerNorm+Conv1d+LayerNorm. The idea of using the first combination comes from reference [57]. Specifically, the FC operation on the input is more likely to have a symmetric, nonsparse distribution, that is, "more Gaussian" [62], and the subsequent BN is likely to generate activations with stable distribution. A similar structure can be seen in other models [63], [64]. The purpose of the second combination operation is to standardize the channel layer features of the input data and increase the diversity of features through the convolution operation. In general, there are two goals by applying two combinations between the channel attention module and the dynamic self-aggregation block: one is to standardize the data, accelerate network training, and alleviate overfitting; the other is to increase the number and diversity of features to enhance the classification performance of the network.

E. Loss Function

An appropriate objective function is critical for optimizing the proposed model. The focal loss [65] can effectively alleviate the impact of category imbalance in classification. Here, we adopt the focal loss as loss function of ADSN. The focal loss is expressed by

$$\text{Loss}_{\text{FL}} = -\sum_{m=1}^{M}\sum_{l=1}^{L}\boldsymbol{y}_l^m\left(1-\hat{\boldsymbol{y}}_l^m\right)^{\gamma}\log\left(\hat{\boldsymbol{y}}_l^m\right) \quad (14)$$

where $\boldsymbol{y}$ and $\hat{\boldsymbol{y}}$ are the true label and the predicted label, respectively, M is the number of samples in batch, L is the number of classes, and the focusing parameter is $\gamma = 1$. Notably, the weight parameters of the model are updated through backpropagation and adaptive moment estimation.

III. Experiments and Analysis

A. Study Area and Dataset

To evaluate the performance of proposed models, the datasets considered are publicly available Sentinel-2 and Landsat time series multispectral images, as well as the land cover reference, as listed in Table I.

Dataset-1: The first dataset came from an open-access dataset on Sentinel-2 tile (T31TFM) for crop classification [27], which is located in Southern France [see Fig. 4(a)]. The acquired Sentinel-2 A images span 24 dates (T) from January to October 2017. Each image contains ten bands (C) of top-of-canopy reflectance, excluding coastal aerosol, water vapor, and shortwave infrared (SWIR) cirrus. To unify the spatial resolution between bands, all 20-m bands are resampled to 10 m. The Orfeo Toolbox [66] is used to linearly interpolate the values of cloudy pixels from the first previous and next available pixels. This dataset is extremely challenging due to the diverse crop types and imbalanced class labels. Fifteen types of crops are used for model evaluation, involving a total of 82 061 individual parcels. Each parcel has a corresponding geo-referenced polygon and class label founded on the French Land Parcel Identification System, which is employed to construct SITS. There are two formats at the parcel scale, including patches and pixel sets. For the patch format, SITS of each parcel is resized as a $T \times C \times H \times W$ ($H = 32$, $W = 32$) tensor obtained by interpolating the spectral channels of each date. The pixels outside of the parcels with a fixed size of 32×32 are padded with zeros. In the pixel sets' format, the pixels of each parcel are stored in the tensor of $T \times C \times N$, and the total number of pixels N can be customized. Meanwhile, the geometric features of the parcels need to be calculated to alleviate the loss of spatial structure.

Dataset-2: A series of Landsat analysis ready data (ARD) surface reflectance images from Iowa City, IA, USA, was selected as the second dataset [see Fig. 4(b)]. The target site (900×900 pixels) is included in the footprint of h18v07 according to the ARD system, and the SITS consists of Landsat seven ETM+ and Landsat eight OLI images with 30-m resolution from May 22, 2018, to September 18, 2018. The image of each date contains six spectral channels, including red, green, blue, near-infrared, SWIR-1, and SWIR-2. The pixels marked with shadow, cloudy, unclear, and filled are all abandoned. In addition, any pixels with fewer than seven valid observations from the start date are removed from the dataset. Founded on the nearest valid observations before and after the target time step, temporal linear interpolation was performed to fill the gaps caused by image quality control. Finally, the obtained SITS are composed of 23 time steps with an interval of seven days, and the images are stored as multiple $T \times C \times H \times W$ ($H = 18$, $W = 18$) tensor in the format of the patch, where T is the number of time steps, C is the number of channels, and $H \times W$ is the spatial size of the patch. There are ten types of target label benefited from the cropland data layer (CDL), which is supported by the United States Department of Agriculture. Note that "others" refer to the merged class of other land covers/land use. Similarly,

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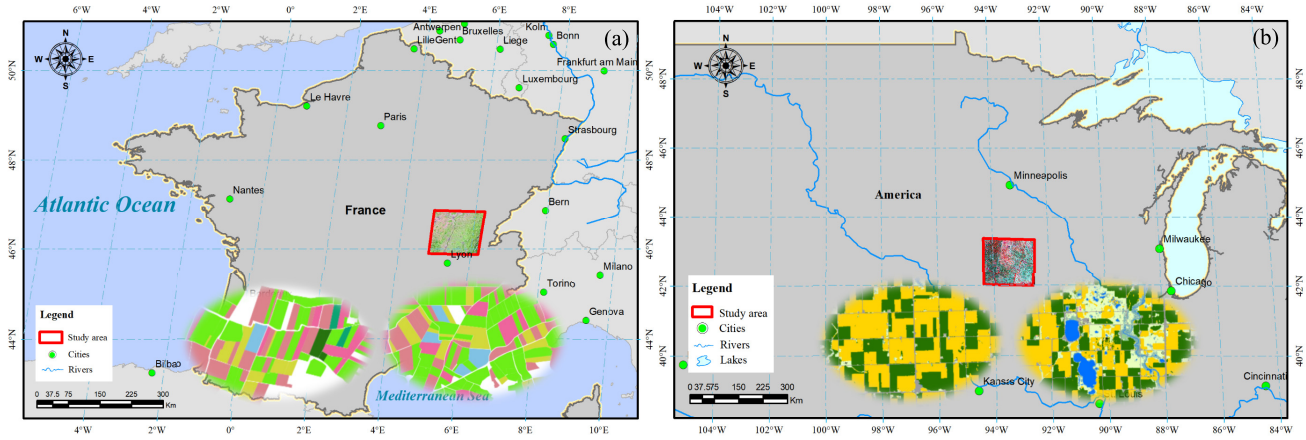


Fig. 4. Location and examples of ground truth on (a) Dataset-1 and (b) Dataset-2 for SITS classification.

TABLE I
CLASS NAME AND NUMBER OF SAMPLES IN THE TWO DATASETS

Study Area	No.	Class Name	Number of Samples	Proportion of crop-specific(%)
ARA and BFC, France	1	Spring Cereal	562	0.68
	2	Summer Cereal	21474	26.17
	3	Winter Cereal	23010	28.04
	4	Sorgho/Willet/Moha	209	0.25
	5	Cereal	729	0.89
	6	Leguminous Fodder	2729	3.33
	7	Winter Rapeseed	4091	4.99
	8	Sunflowers	665	0.81
	9	Soy	3060	3.73
	10	Protein Crops	361	0.44
	11	Potato	291	0.35
	12	Fruits/Legumes/Flowers	801	0.98
	13	Ligneous	462	0.56
	14	Grapevine	18466	22.50
	15	Non Agricultural	5151	6.28
	Total		82061	100
Iowa, America	1	Others	792	0.10
	2	Corn	393545	48.59
	3	Soybeans	261186	32.25
	4	Alfalfa	1562	0.19
	5	Open Water	7573	0.93
	6	Developed Space	54356	6.71
	7	Barren	437	0.05
	8	Deciduous Forest	7299	0.90
	9	Grassland/Pasture	63525	7.84
	10	Wetlands	19725	2.44
	Total		810000	100

Note: ARA:Auvergne-Rhone-Alpes, BFC:Bourgogne-Franche-Comte.

severely unbalanced samples are also a prominent challenge for the constructed dataset.

B. Experimental Configuration

All the experiments are run on a computer equipped with a ten-core Intel Core i9-9900X CPU at 3.5 GHz (128-GB RAM) and NVIDIA GeForce RTX 2080Ti GPU (11-GB RAM). The software environment is 64-b Windows 10, and the different architectures presented here are implemented in the deep learning framework PyTorch.

All deep learning models were trained by the focal loss [65] and Adam optimizer [67] with the learning rate of 0.001. Meanwhile, the fivefold cross-validation scheme is implemented in the training process of all networks, and the dataset is divided into training, validation, and test sets with a ratio of 3:1:1 in each fold. The function of the validation set is to filter the best model parameters obtained from training, and then, the quality of the parameters is evaluated through a test set. In our experiment, 100 epochs are set to complete the convergence of the model. In addition, an early termination mechanism is designed to prevent the model from overfitting. The criterion for the stop of model training is that the number of epochs is greater than 50, and the difference between the highest and lowest training accuracy values in the last six epochs is less than 0.01.

Taking into account the challenge of the strong imbalance in the class proportions of datasets, the quantitative evaluation metrics used are class-specific accuracy, overall accuracy (OA), kappa coefficient (κ), mean per-class Intersect over Union (mIoU), and mean per-class F1 score (mF1).

C. Parameter Setting

This section mainly introduces the specific settings on the ADSN parameters. To balance the limitation of computer memory and the speed of model training, the batch sizes of Dataset-1 and Dataset-2 are determined to be 128 and 512, respectively. According to the research of Garnot *et al.* [27], the general parameters of multihead attention are set, including $d_k = 32$, $d_v = 64$, $d_{\text{model}} = 128$, $d_{\text{inner}} = 256$, $n_{\text{head}} = 4$, and $n_{\text{layer}} = 1$. The distance range is a key factor that affects the representation of the relative position, where the maximum absolute value m is assigned as 20. The kernel size of dynamic convolution determines the representation capacity and receptive field, which indirectly affects the classification accuracy of the model, and a dual-kernel configuration is used here. For the Dataset-1, the dual-kernel size of dynamic convolution is set to $\{3, 15\}$, while it is set to $\{3, 17\}$ for the Dataset-2.

D. Comparison With the State of the Arts

The proposed ADSN is compared with RF [68] and seven deep learning models: PSE-TAE [27], Transformer [2], CNN [34], TempCNN [30], TCN [69], StarRNN [28], and Bidirectional LSTM (Bi-LSTM) [70], and we have reimplemented these models on datasets. More specifically, the first two deep learning benchmarks are attention-based methods; the latter three models are based on convolution; and StarRNN and LSTM use recurrence as the basis.

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TABLE II

CANDIDATE PARAMETERS OF DEEP LEARNING MODELS FROM COMPETITORS FOR TUNING

Model	Candidate parameters	Reference
PSE-TAE	**n_{head}: {2, 4 , 6, 8}, d_k: {32, 64},** **mlp_3: {[256, 128, 128], [512, 128, 128], [768, 128, 128], [1024, 128, 128]}** mlp_1: [10, 32, 64] (Dataset-1) / [6, 32, 64] (Dataset-2), mlp_2: [132, 128] mlp_4: [128, 64, 32, 15] (Dataset-1) / [128, 64, 32, 10] (Dataset-2)	[27]
Transformer	**n_{head}: {2, 4, 6, 8}, dk: {16, 32, 64}, dv: {32, 64}**, d_{model}: 128, d_{inner}: 256, n_{layer}: 1	[2]
CNN	**$kernel_{size}$: {3, 5, 7, 9}**, global average pooling, number of kernels: [32, 64, 96]	[34]
TempCNN	**$kernel_{size}$: {3, 5, 7, 9}**, **$hidden_{dims}$: {64, 128}**	[30]
TCN	**$kernel_{size}$: {3, 5, 7, 9} num_{layers}: {4, 6, 8}**, $hidden_{dims}$: 64	[31]
StarRNN	**$hidden_{dims}$: {64, 128} num_{layers}: {4, 6, 8}**	[28]
Bi-LSTM	**$hidden_{dims}$: {64, 128} num_{layers}: {4, 6, 8}**	[70]

TABLE III

OPTIMAL PARAMETER VALUES OF DIFFERENT COMPETITORS FOUNDED ON DATASET-1 AND DATASET-2

Model	Optimal parameter value on Dataset-1	Optimal parameter value on Dataset-2
PSE-TAE	n_{head}: 8, d_k: 32, mlp_1: [10, 32, 64], mlp_2: [132, 128], mlp_3: [1024, 128, 128], mlp_4: [128, 64, 32, 15]	n_{head}: 8, d_k: 64, mlp_1: [6, 32, 64], mlp_2: [132, 128], mlp_3: [1024, 128, 128], mlp_4: [128, 64, 32, 10]
Transformer	n_{head}: 8, d_k: 16, d_v: 64, d_{model}: 128, d_{inner}: 256, n_{layer}: 1	n_{head}: 8, dk: 16, dv: 32, d_{model}: 128, d_{inner}: 256, n_{layer}: 1
CNN	$kernel_{size}$: 9, global average pooling, number of kernels: [32, 64, 96]	$kernel_{size}$: 3, global average pooling number of kernels: [32, 64, 96]
TempCNN	$kernel_{size}$: 5, $hidden_{dims}$: 64	$kernel_{size}$: 3, $hidden_{dims}$: 128
TCN	$kernel_{size}$: 7, num_{layers}: 4, $hidden_{dims}$: 64	$kernel_{size}$: 7, num_{layers}: 4, $hidden_{dims}$: 64
StarRNN	$hidden_{dims}$: 128, num_{layers}: 4	$hidden_{dims}$: 128, num_{layers}: 4
Bi-LSTM	$hidden_{dims}$: 128, num_{layers}: 8	$hidden_{dims}$: 128, num_{layers}: 4

TABLE IV

CLASSIFICATION ACCURACIES OF DATASET-1 THROUGH DIFFERENT METHODS

Class	RF	PSE-TAE	Transformer	CNN	TempCNN	TCN	StarRNN	Bi-LSTM	ADSN
OA(%)	89.44	94.39	92.80	94.00	93.61	92.88	92.28	92.67	**94.56**
$\kappa \times 100$	86.45	92.91	90.91	92.42	91.90	90.97	90.25	90.72	**93.12**
mIoU(%)	45.74	**66.44**	60.91	64.08	61.25	52.92	58.74	58.92	66.17
mF1(%)	54.42	**76.67**	71.18	74.23	71.39	62.17	69.29	69.25	76.01
Spring Cereal	67.24	72.53	58.64	74.08	52.72	61.97	48.60	47.32	64.93
Summer Cereal	92.05	97.45	97.48	97.06	97.01	96.73	96.64	96.51	97.50
Winter Cereal	90.97	96.53	96.26	96.14	95.84	94.94	96.13	95.39	96.53
Sorgho/Willet/Moha	-	55.19	28.46	55.66	60.00	48.00	32.08	41.98	52.42
Cereal	-	40.11	32.28	42.56	46.47	37.23	34.93	40.18	39.90
Leguminous Fodder	80.50	86.69	83.46	88.71	85.45	83.83	79.33	82.73	89.99
Winter Rapeseed	98.05	98.61	98.51	98.87	98.65	96.93	98.11	98.08	98.82
Sunflowers	92.86	91.52	92.18	93.94	91.57	62.02	84.28	90.76	94.74
Soy	92.38	95.19	93.97	95.42	94.65	92.71	92.48	94.58	95.22
Protein Crops	84.62	58.51	54.72	67.66	76.10	62.24	38.72	54.23	61.24
Potato	92.31	59.72	50.77	65.58	58.33	53.13	52.08	56.76	57.87
Fruits/Legumes/Flowers	77.19	74.90	63.57	79.85	77.54	68.58	68.68	68.72	75.93
Ligneous	65.45	73.95	71.13	76.26	67.29	75.86	55.80	70.86	68.48
Grapevine	88.43	97.01	96.03	95.93	95.49	95.34	93.86	95.43	96.32
Non Agricultural	67.65	80.74	69.70	77.21	75.38	73.11	78.43	71.43	82.85

Note: '-' means no value, that is, the model fails to classify corresponding category.

To make the obtained results more convincing and comparable, we have tuned all the competitors in detail on the two datasets. The usage of Dataset-1 mainly refers to the literature [27], and CNN adopts the patch format, while the remaining models adopt pixel sets. There are two purposes to do this: one is to further verify the advantages of pixel sets, and the other is to increase the diversity of contrasts and make the experimental results more convincing. According to the characteristics of each model, we select the relevant parameters that affect the classification accuracy to tune for finding the best parameter value. RF is implemented through the scikit-learn framework [71]. The RF optimal parameters obtained are {number of trees: 500, maximum depth of trees: 50 (Dataset-1) or 25 (Dataset-2), max number of features considered for splitting a node: square root calculation, criterion of a split: Gini impurity}. The specific candidate parameters of deep learning models from competitors are shown in Table II. The parameters in bold are the parts that need to be tuned, which have a greater impact on the classification results, and the rest of the parameters are set according to the reference. For the meaning of related parameters, please refer to the references of Table II. The candidate parameters of each model are traversed and combined for training, and the optimal parameter values of different competitors are selected based on the classification accuracy (see Table III). It is obvious that this process is time-consuming and necessary.

1) Quantitative Evaluation: Tables IV and V demonstrate the metrics of different methods. Most deep learning models yield comparable or better results than RF. This indicates that deep learning can learn hierarchical and discriminative

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TABLE V
CLASSIFICATION ACCURACIES OF DATASET-2 THROUGH DIFFERENT METHODS

Class	RF	PSE-TAE	Transformer	CNN	TempCNN	TCN	StarRNN	Bi-LSTM	ADSN
OA(%)	91.34	91.37	90.61	91.60	**92.60**	91.51	89.94	90.40	91.45
$\kappa \times 100$	86.54	86.56	85.39	86.93	**88.51**	86.79	84.36	85.08	86.73
mIoU(%)	49.42	49.86	47.13	47.76	50.24	44.46	45.83	46.77	**50.33**
mF1(%)	61.96	62.75	59.57	60.04	61.97	55.21	58.46	59.23	**63.20**
Others	69.14	34.29	35.04	26.54	36.82	-	18.46	33.91	39.16
Corn	96.09	96.21	95.61	96.33	96.97	96.24	95.60	95.85	96.48
Soybeans	96.17	96.18	95.98	95.96	96.78	96.17	95.26	95.59	96.16
Alfalfa	75.05	62.41	61.68	67.61	67.63	64.34	48.54	62.74	62.09
Open Water	72.33	68.44	64.67	69.60	65.92	69.84	58.05	58.59	68.03
Developed Space	71.08	68.66	65.65	73.12	77.27	70.21	60.27	66.09	67.44
Barren	88.24	41.63	27.15	40.85	-	44.74	18.35	39.13	41.04
Deciduous Forest	65.75	58.18	58.92	62.93	59.86	54.81	49.13	57.27	58.11
Grassland/Pasture	70.57	75.29	72.37	73.12	76.70	75.18	77.35	71.21	76.76
Wetlands	65.27	62.16	62.34	61.73	61.47	60.60	65.52	58.90	63.74

Note: '-' means no value, that is, the model fails to classify corresponding category.

high-level features, which are useful for SITS classification. The unsatisfactory accuracy of Bi-LSTM and StarRNN is probably caused by the underlearning of discriminative features. The proposed ADSN achieves competitive classification accuracy in comparison with other baselines. Specifically, the proposed network results in the average accuracy increments of 1.80% and 0.28% for OA and 2.30% and 0.46% for κ on each dataset. Moreover, the ADSN outperforms other deep learning networks in mIoU by about 5.70% and 2.89% points and mF1 by approximately 5.41% and 3.60% points on two datasets. Similarly, PSE-TAE performs well on Dataset-1, producing the highest mIoU and mF1. It is a powerful candidate classifier in practical applications. The mIoU and mF1 of Transformer and TCN are relatively low. Due to its competitive OA and κ, TempCNN shows good performance on Dataset-2. Notably, TempCNN does not successfully classify Barren, and its performance may be affected by the imbalance of categories from datasets. It can be seen that the designed framework makes ADSN robust and generalizable.

2) Qualitative Evaluation: To compare the classification results of different methods more intuitively, the classification maps are shown in Figs. 5 and 6. To make the classification results have a better visualization effect, a partially enlarged classification mapping of Dataset-1 is shown. From the crop classification task of Dataset-1, the classification results acquired by ADSN are more consistent with the reference map, and the salt and pepper effect is weak. Since the OA gap of different models is small, and the maximum gap is 2.66%, the visual difference of the classification maps of various models on Dataset-2 is not obvious. In addition, the developed space is an aggregation of open space, low-intensity space, and high-intensity space; this mixed scene causes confusion with other surrounding land types since interference from noise and scale effects. Overall, the ADSN provides a good visual sense of classification, and the high-precision mapping makes it a powerful alternative for large-scale applications.

E. Effects of Parameter Selection

Spectral–temporal feature learning is a critical part of proposed ADSN training and feature extraction. Therefore, it is necessary to evaluate the impact of hyperparameters from dynamic self-aggregation blocks on classification accuracy.

1) Maximum Absolute Value m: From (11), the representation of relative position is influenced by distance range, which is decided by m. Here, a set of different m {6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26} is considered to find the optimal distance for the model on each dataset. The choice of parameters mainly considers two factors: one is whether the classification accuracy of the model is satisfactory, and the other is the consistency of parameter selection and keeping the model with an appropriate number of parameters. As shown in Fig. 7, the classification accuracy of ADSN on two datasets reaches the ideal state when $m = 20$. Although the classification accuracy is similar on Dataset-1 when the value of m is 20 and 22, respectively, the number of parameters of the model becomes larger when $m = 22$, which will increase the computational cost. Therefore, the optimal value of m is 20.

2) Kernel Sizes of Dynamic Convolution: The kernel size determines the representation capacity and receptive field that indirectly affects the classification accuracy. The kernel sizes of $\{3, 5, 7, 9, 11, 13, 15, 17\}$ are included in our consideration. In order to reduce the amount of calculation and promote a multiscale effect, the collocation of dual-kernel is applied in dynamic convolution. As illustrated in Fig. 8, the ADSN with dual-kernel $\{3, 15\}$ reached best OA on Dataset-1, while the dual-kernel set to $\{3, 17\}$ performs best on Dataset-2.

F. Function of Channel Attention Module

To further clarify the role of the channel attention module in ADSN, here is a visual display of the weights activated by the channel attention module based on Dataset-1 (see Fig. 9). All samples were used to participate in the pretrained ADSN for classification. Specifically, we extracted the weight values generated by the channel attention module, and the mean values of samples were taken as the visualization result of the activation weights, which is the weight matrix (24 time steps $\times$ 10 channels).

It can be seen from Fig. 9 that most of the weight values are distributed between 0.44 and 0.56, and the weights of different

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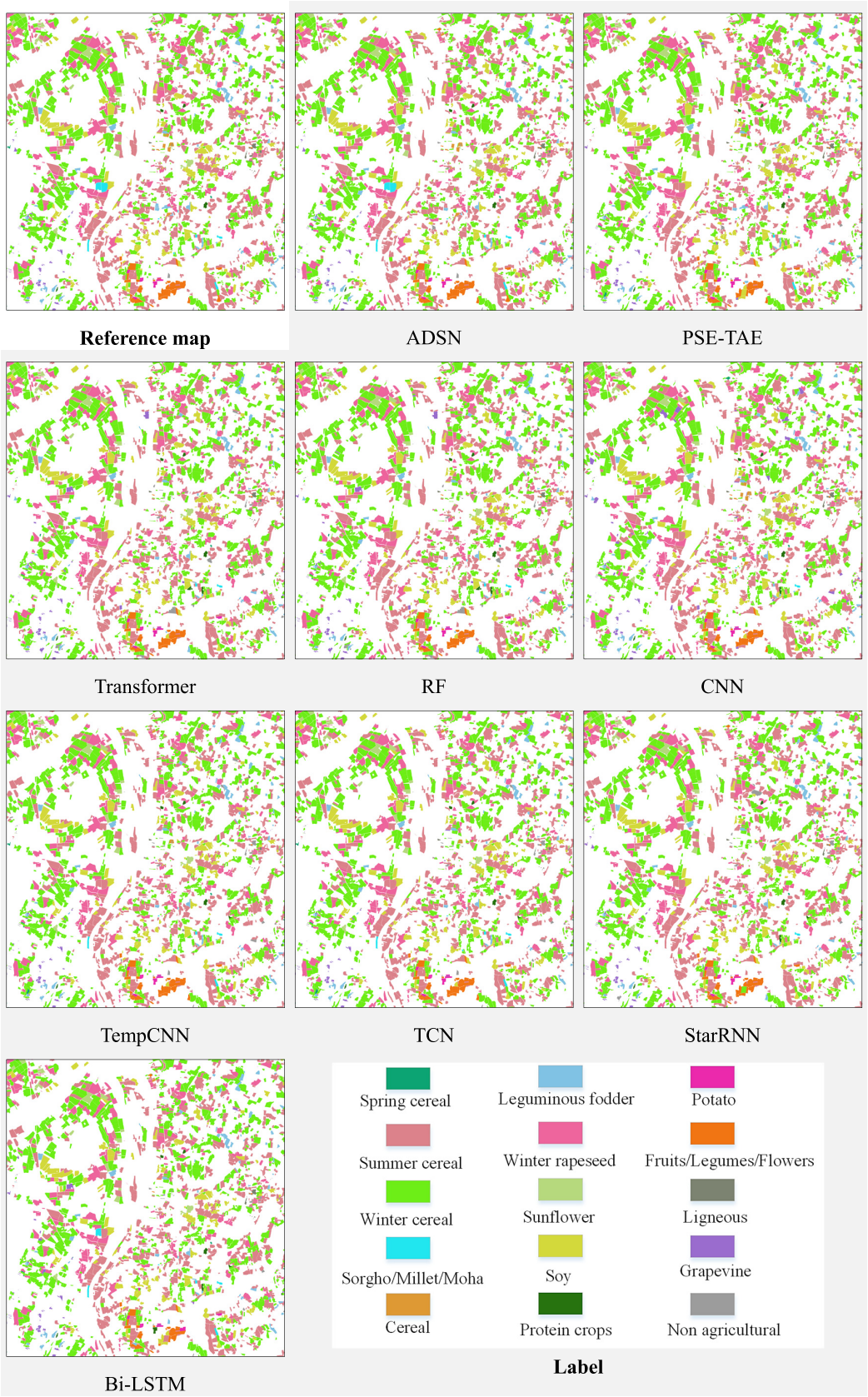


Fig. 5. Partially enlarged classification maps of Dataset-1 through RF and various deep learning networks.

channels corresponding to different time steps are diverse. This shows that the channel attention module adaptively learns the weight of each spectral band in SITS. Based on the channel perspective of the SITS, particularly, the weight values of

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Fig. 6. Classification maps of Dataset-2 through RF and various deep learning networks. The class in red belongs to invalid observations, that is, any pixels with fewer than seven valid observations from the start date are removed from Dataset-2.

channel 5, 6, and 8 are significantly larger than those of other channels. Since these three channels are all vegetation red edge bands, their weights are strengthened in favor of crop classification to improve classification accuracy.

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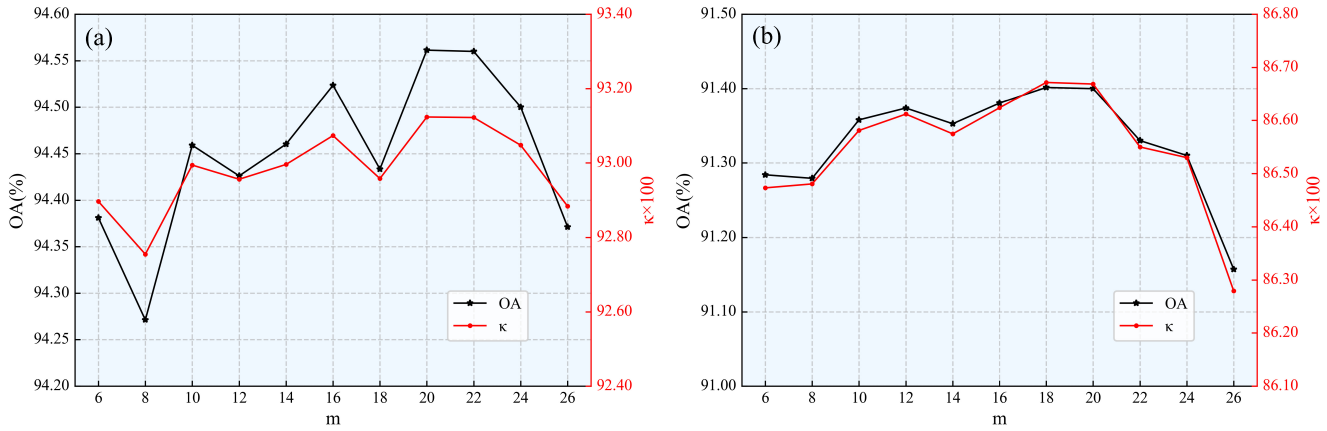


Fig. 7. OA(%) and κ of ADSN with a different maximum absolute value m in (a) Dataset-1 and (b) Dataset-2.

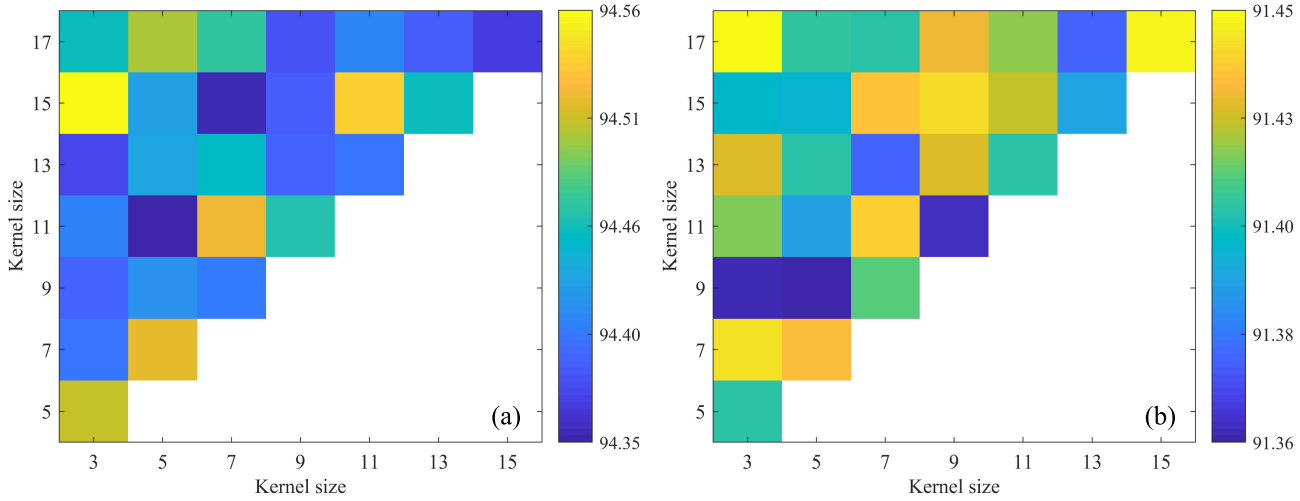


Fig. 8. OA(%) of ADSN with different combinations of dual kernel of dynamic convolution in (a) Dataset-1 and (b) Dataset-2.

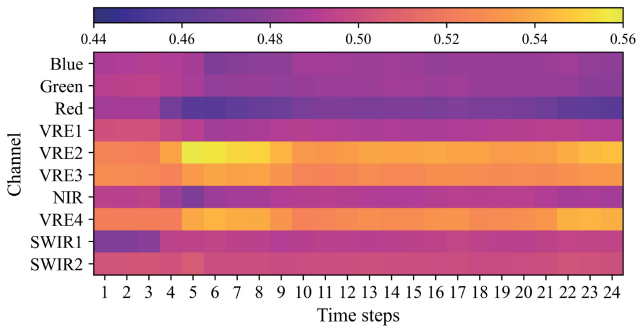


Fig. 9. Weight activation mapping through channel attention module of ADSN based on Dataset-1. VRE: vegetation red edge; NIR: near infrared, and SWIR: shortwave infrared.

G. Ablation Study

1) Accuracy Response Analysis: To analyze the contribution of different ingredients of the proposed model, ADSN is compared with several models lacking key modules, including ADSN only with dynamic self-aggregation block (DSN), ADSN without RPE multihead attention (AWR), and ADSN without dynamic convolution (AWD). The results are shown in Table VI. On the whole, the classification accuracy of ADSN is better than others. The AWD has the worst classification effect, which indicates that dynamic convolution is an important part of the construction of ADSN. Furthermore, the ranking of the contribution of different modules can be summarized as dynamic convolution > RPE multihead attention > channel attention module.

2) Visual Analysis of Feature Separability: Deep learning network helps to better condense and express complex features through the cascaded layer structure [72]. In this part, we visualized deep hidden features of ADSN and variants, which aims to analyze the separability of features and the impact of different modules on them. Dataset-1 is used for this analysis. Based on the pretrained ADSN, DSN, AWR, and AWD models, we extracted the 128-D hidden feature vectors after the two modules, including the channel attention module and the dynamic self-aggregation block. Then, the t-distributed stochastic neighbor embedding (t-SNE) [73] was applied to project them into a 2-D embedding space, as shown in Fig. 10. In order to have a better visual effect, 200 samples

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TABLE VI
QUANTITATIVE RESULTS OF ADSN AND ITS VARIANTS WITHOUT KEY MODULES

	Channel attention module	PRE multi-head	Dynamic convolution	OA(%)	$\kappa \times 100$	mIoU(%)	mF1(%)
ADSN	√	√	√	94.56	93.12	66.17	76.01
				91.45	86.73	50.33	63.20
DSN	×	√	√	94.35	92.86	65.61	75.71
				91.43	86.70	50.46	63.41
AWR	√	×	√	94.46	92.99	65.55	75.56
				91.32	86.53	49.78	62.58
AWD	√	√	×	94.34	92.84	65.52	75.60
				91.10	86.20	49.39	62.19

Note: Light pink means Dataset-1, light blue denotes Dataset-2.

TABLE VII
NUMBER OF PARAMETERS ON DIFFERENT DEEP LEARNING MODELS

Parameters	PSE-TAE	Transformer	CNN	TempCNN	TCN	StarRNN	Bi-LSTM	ADSN
Dataset-1	312047	249331	726093	442575	207887	71587	2928675	321093
Dataset-2	564522	180902	178228	1615754	205514	69910	1329174	311400

TABLE VIII
RUNNING TIME OF DIFFERENT DEEP LEARNING MODELS

Time(s/epoch)	PSE-TAE	Transformer	CNN	TempCNN	TCN	StarRNN	Bi-LSTM	ADSN
Dataset-1	50.82	48.14	92.33	47.09	47.85	51.28	86.25	51.31
Dataset-2	36.11	25.89	41.18	35.94	32.47	39.64	47.76	33.89

for each of the 15 classes are selected for embedded feature characterization. The clusters of each category are marked with different colors. One thing that is easy to find is that deeper network layers make features more separable. Compared with the DSN, AWR, and AWD, the distinct separable manifolds of ADSN are obvious for various land use classes. Specifically, taking Winter Rapeseed as an example, the only one sample of Protein Crops in ADSN is confused with Winter Rapeseed in terms of embedded features obtained after the dynamic self-aggregation block, while multiple different categories in the other three models produce confusion with Winter Rapeseed, such as potato and nonagricultural. Moreover, the dynamic self-aggregation block plays a key role for ADSN. In the AWD, the separability of embedded features is worse than that of the other three models. This performance indirectly proves this point. In general, the channel attention module and the dynamic self-aggregation block enhance the expression and separability of features, and they are the core components of ADSN.

H. Analysis of Stability and Complexity

1) Stability Analysis: The underfitting or overfitting will interfere with the performance of deep learning models. To this end, we evaluated the training loss, training accuracy, validation accuracy, and mIoU of model training within 100 epochs. As shown in Fig. 11, compared with the attention-based Transformer, the convolution-based TempCNN, and the recurrence-based Bi-LSTM, the ADSN not only owns a faster convergence speed but also holds a better fitting ability. Besides, in order to test the stability of different models, we performed statistics on the OA and mIoU of all models during the fivefold cross-validation process (see Fig. 12). The statistical indicators include maximum, minimum, median, average, first quartile, and third quartile. The classification performance of RF on Dataset-1 is poor except for Dataset-2. It can be seen that the fluctuation range of TCN, CNN, TempCNN, and Bi-LSTM on OA is relatively larger than that of other models; the deep learning models with stable performance are PSE-TAE, Transformer, and ADSN; and the fluctuation of mIoU of TCN and CNN is particularly obvious. Overall, the proposed ADSN with small fluctuations and ideal precision especially highlights its stability.

2) Computational Complexity: Table VII shows the number of network parameters in all deep learning models. Due to the difference in the length of the time series and the number of categories between the two datasets, it is necessary to fine-tune the fully connected layer while maintaining the overall structure of the model unchanged. Different deep learning models are tuned based on Dataset-1 and Dataset-2. Therefore, the total amount of model parameters for different datasets is diverse. Specifically, Bi-LSTM, TempCNN, and CNN have more parameters than other models. In contrast, the proposed ADSN possesses comparable parameters to other lightweight models while maintaining competitive classification accuracy. Besides, the model training speed is relatively slow and time-consuming when the amount of input data is large. As shown in Table VIII, we have summarized the model training time of one epoch in the training phase, and the statistical result is the average of the running time of ten epochs. It can be seen that each model takes more time to complete an epoch training on Dataset-1 than on Dataset-2. In general, the training time of Bi-LSTM and CNN is longer than other models. Since TempCNN

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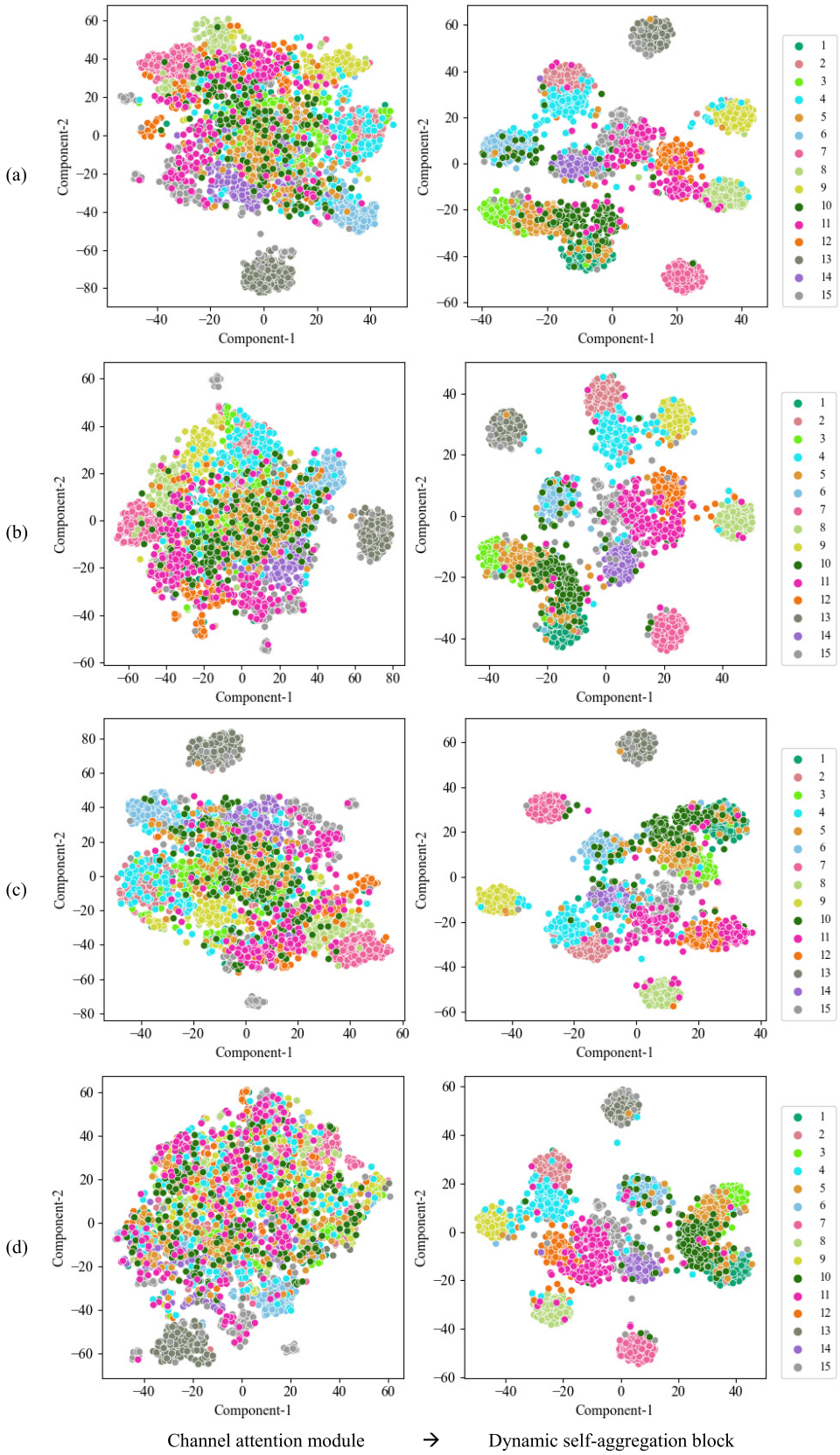


Fig. 10. Feature embedding at different depths of (a) ADSN, (b) DSN, (c) AWR, and (d) AWD based on Dataset-1. The hidden features are recorded from behind the channel attention module and dynamic self-aggregation block, respectively. The corresponding class names of the class numbers have been listed in Table I.

uses temporal convolution architecture, its training speed is fast despite a large number of parameters, which is confirmed in the literature [74]. Although ADSN has no obvious advantages in training time, the bottleneck of model training

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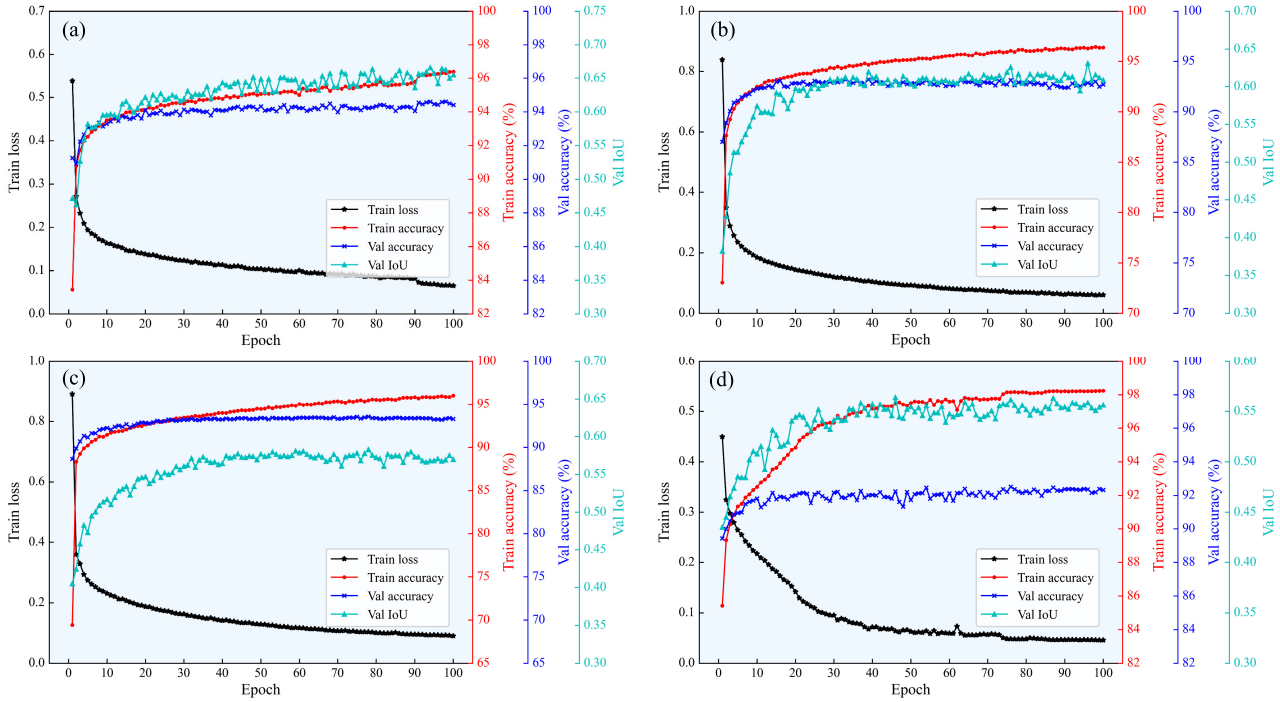


Fig. 11. Training loss, training accuracy, validation accuracy, and mIoU of model training on (a) ADSN, (b) Transformer, (c) TempCNN, and (d) Bi-LSTM within 100 epochs.

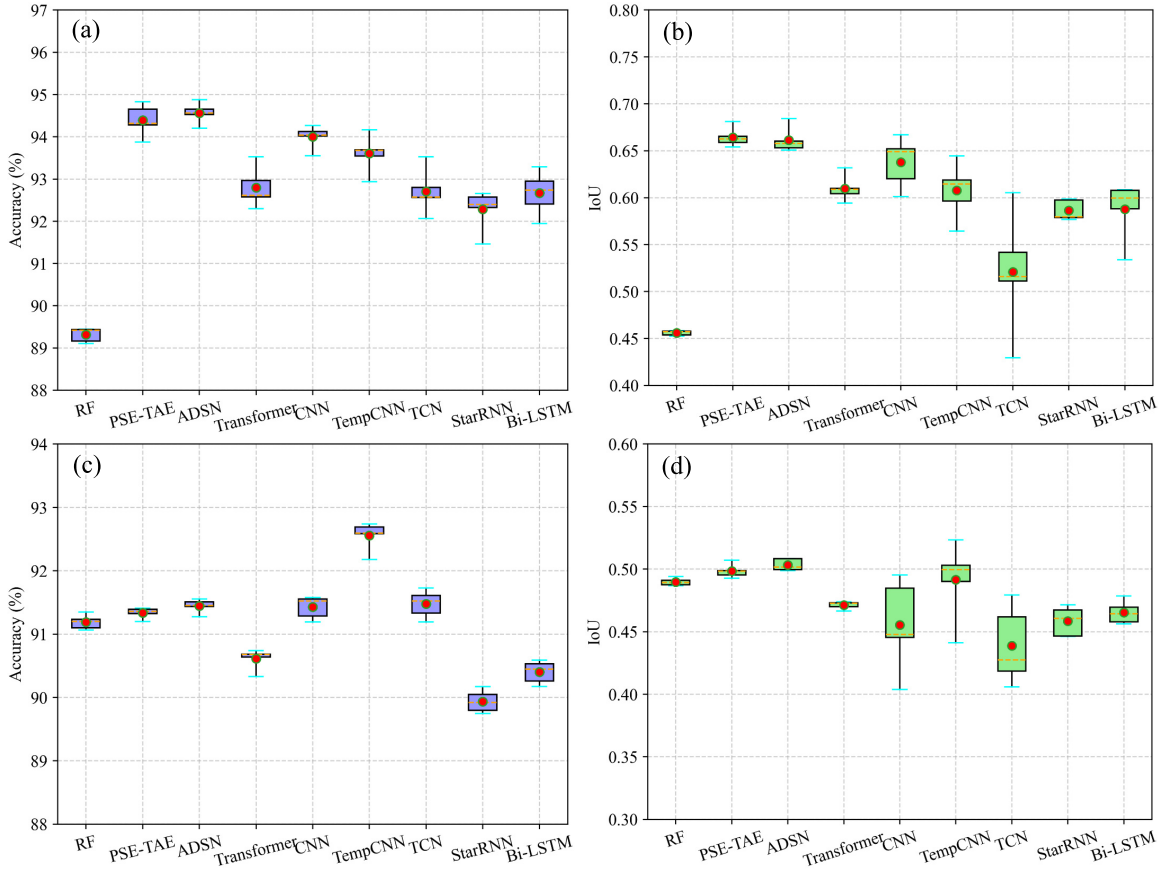


Fig. 12. Statistics on OA [(a) Dataset-1 and (c) Dataset-2] and mIoU [(b) Dataset-1 and (d) Dataset-2] of different models in the fivefold cross-validation process.

speed will be greatly weakened with the popularization of multicore computing, parallel computing, and cloud computing technology. Currently, what needs to be improved for ADSN is to further reduce the number of parameters.

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IV. Conclusion

To fully explore the high-precision classification of SITS, we proposed a new deep learning framework ADSN by enhancing the adaptive representation of spectral and temporal features.

The joint use of the channel attention module and the dynamic self-aggregation block is the key to improve ADSN productivity. The former filters valuable bands, and the latter achieves powerful capture of time-dimensional structural features by combining the PRE multihead attention and dynamic convolution. After extensive evaluation and validation on two SITS datasets under extremely uneven samples, the ADSN achieves state-of-the-art or comparable classification results compared with other baselines, and the optimal OA is all higher than 91%. In addition, accurate land cover mapping of different regions shows that the model framework has a certain generalization ability, and it can be used as an effective alternative for time series scenarios, such as target extraction. Furthermore, the proposed network is demonstrated to have a good classification effect and stability by investigating the sensitivity of its parameters and modules.

The future direction of our work is to further lighten the model, aiming to make it better suitable for large-scale and multisource images (e.g., SAR, nighttime light, and unmanned aerial vehicle images) applications. By exploiting Google Earth Engine, the framework developed here is also expected to serve as a template to maximize the potentials of SITS for monitoring the Earth's surface.

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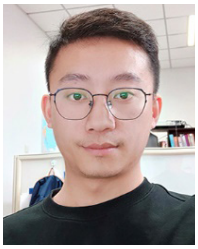
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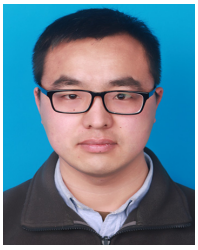
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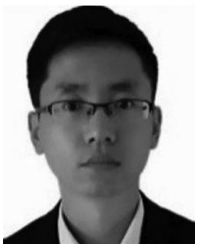
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